THE COMPLETE GUIDE · SMART INDIA HACKATHON SIH26047 · PATIENT CASE-TAKING SOFTWARE

OPD Saathi

A multilingual check-in kiosk, phone check-in and live queue for crowded government OPDs: the patient's history is ready, in English, before they reach the doctor.

Everything in one place: what the system does, how a visit flows through it, every question it asks, the OPD card, every screen and who uses it, the AI and its limits, the technology, how it was tested, what was built when, and how to run the demo.

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19 Limits and words used

Hindi · Marathi · English | Kiosk or the patient's own phone | Live queue + QR status | Fair, explainable triage | OPD card for returning patients | AI drafts with sources, never fact | Old papers read as drafts | Doctor's own notes | Two-step sign-in · audit log | History only — no diagnosis

25 October 2026 · version 0.7 with Phase 7 G3, the logo and the per-day reports (`main` , pull request #74). Screenshots come from the application with fictional patients; older ones show the earlier logo. Source: `docs/complete-guide/guide.html` .

START HERE

The system on one page

Government OPDs see hundreds of patients a morning. Patients speak different languages, describe problems in their own words and wait without knowing their turn; doctors get a few minutes each and spend much of it collecting basic history.

What OPD Saathi does

- Collects the patient's history **before** the consultation, in their language, at a kiosk or on their own phone, by tapping, speaking or typing.
- Spots danger signs first and gives every patient a **fair, explained place** in the queue.
- Gives the doctor an **English case sheet** with the patient's own words, earlier visits and old test reports beside it.
- Shows staff a **live queue** and patients a **live status page** on their phone.
- Gives administrators reports, an audit log and control of staff, kiosks, departments and features.

What it deliberately does not do

- It never diagnoses, names a disease, suggests a medicine or dose, or checks drug interactions.
- AI only drafts and double-checks: every AI line is labelled, cites the patient's answers and is accepted or rejected by a doctor. AI can raise caution, never lower it.
- Patients never see a score; staff stay responsible for who is seen first.
- Clinical wording and translations are drafts until clinicians and native speakers sign them off.

One patient's journey

Check-in — Kiosk or own phone: language, consent, danger signs, 8–15 questions → **Token + QR** — Department, place in line, OPD card for next time → **Nurse** — Live queue, alerts, fair calling order, missed calls → **Doctor** — Case sheet, AI draft to check, old papers, own notes, PDF → **Admin** — Reports, settings, audit log, system health

Who uses what

WHO	WHERE	WHAT THEY DO THERE
Patient	Kiosk, or a QR poster on their own phone	Chooses Hindi, Marathi or English; agrees; says who is answering; gives age and sex; uses an OPD card if they have one; flags danger signs; answers; adds photos of old papers; confirms the read-back; gets a token, QR and OPD card.
Patient's phone	QR on the token	Live place in line, wait range and progress in their language; can stop AI help or SMS; reads the privacy notice. No app, no login.
Nurse	Queue, Patients	Watches the queue and alerts, calls and recalls patients, confirms danger signs and departments, links returning patients, re-issues lost OPD cards, checks in patients who need help.
Doctor	Queue, Consultations	Calls the next patient, checks the AI draft against its sources, reads earlier visits and old papers, corrects the history, writes their own notes, prints the PDF.
Administrator	Reports, Settings, Audit log, System	Runs staff accounts, kiosks, phone QRs, departments and hospital-wide switches; reads reports and the audit log; checks the installation's health.

3
languages everywhere
Kiosk, phone, status page, staff screens, notices and printed forms in Hindi, Marathi and English.

33
questions in 16 topics
Each patient sees 8–15 of them, chosen by the complaint, after six danger-sign buttons (page 5).

0
diagnoses
History-taking and the queue only, on every screen, in every output and in every AI answer.

HOW IT WORKS

How a visit works, end to end

What the patient and staff see at each step, and what the system does behind the screen. Every step works on the hospital's own server; AI and SMS are optional extras.

STEP	WHAT THE PERSON SEES AND DOES	WHAT THE SYSTEM DOES
1 **Arrive**	Walks up to a kiosk, or scans the waiting-hall QR poster with their own phone.	The kiosk was paired once with a one-time activation code and signs itself in. The phone QR is rate-limited (6 check-ins per phone and 150 per poster in 10 minutes) and can be replaced at once.
2 **Consent**	Chooses Hindi, Marathi or English; hears the care notice and agrees; says who is answering; decides separately about online AI help (and SMS, when the hospital has it).	Nothing is collected before consent. Each purpose (care, AI, SMS) is its own record, audited and withdrawable.
3 **Details**	Age and sex; a phone number is optional. A returning patient types the OPD number and PIN from their last slip.	The phone number is encrypted. The OPD card must fit today's age and sex; wrong tries are limited and audited; the visit is linked to the patient's record.
4 **Safety**	Six danger-sign buttons come before any question.	One tap marks the visit urgent and pushes an alert to every staff board within a second (Redis publish, then WebSocket).
5 **Interview**	8–15 questions, each answered by a tap, by speaking or by typing.	The question engine picks topics from the complaint (keywords, at most two) and applies age and sex rules. Speech goes to Sarvam. 15 red-flag rules check every answer. With AI consent, Gemini may add up to three labelled follow-ups.
6 **Review**	Hears the history read back, changes any answer, adds photos of old papers, confirms.	The read-back comes from fixed templates, never a language model. The English case sheet is built from the bilingual question bank; AI translates free text only with consent. Photos are stored encrypted.
7 **Token**	Gets the token (MED-016), department, place in line, a QR for the phone and the OPD card.	Rules compute the triage score (0–100) and the department. Tokens are numbered per department and day. Urgent patients go first; others get a head start of up to 45 minutes.
8 **Waiting**	The phone shows people ahead and a wait range, and vibrates when the patient is called.	Every queue change is written to an outbox in the database; the worker relays it to Redis every 2 seconds, and boards and status pages update live.
9 **Nurse**	Sees the live board, acknowledges alerts, confirms danger signs, calls and recalls patients.	A call reserves the patient for one doctor; a missed call gets a grace period before "absent". Every view and change goes to the append-only audit log.
10 **Doctor**	Opens the case sheet, checks the AI draft against its sources, reads earlier visits and papers, writes notes, prints the PDF.	Case sheets and notes are versioned, so two people cannot overwrite each other. The output guard blocks any AI text that diagnoses or advises. WeasyPrint makes the PDF.
11 **Reports**	The administrator sees visits, waits, urgency, languages and how often the kiosk history was right.	The worker recounts today and yesterday every 5 minutes and every day at 03:07. Yesterday's unfinished visits close themselves after midnight.

No internet?

The kiosk keeps an encrypted short form and sends it once the network is back; the history still builds from rules.

No AI?

Tapping, typing, recorded voice prompts, the rules engine and the bilingual question bank do the whole visit (page 13).

Server down?

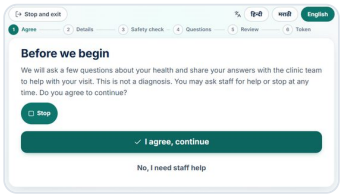
The printed paper fallback pack (forms in three languages, a token roll) keeps the OPD going.

The patient interview, screen by screen

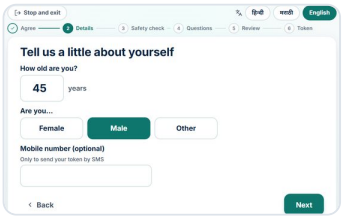
A touch-screen kiosk (any tablet or PC with Chrome or Edge) or the patient's own phone. Six steps show at the top of every screen: Agree → Details → Safety check → Questions → Review → Token.



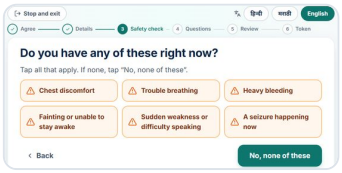
1 Language. Hindi, Marathi or English; every screen can be read aloud.



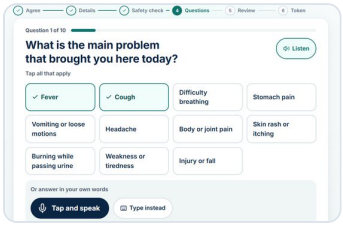
2 Consent. Agree, or "No, I need staff help". Next: who is answering, and the separate AI question.



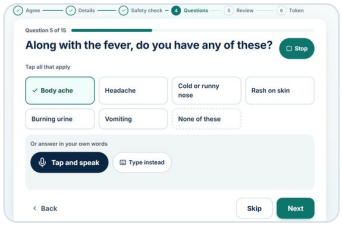
3 Details. Age and sex; a mobile number only for the token SMS. Returning patients use their OPD card here.



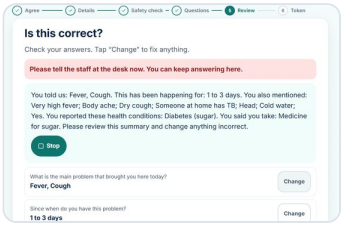
4 Safety check. Six danger signs, or "No, none of these". One tap alerts the staff.



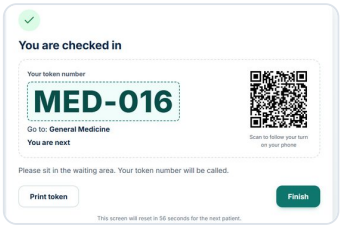
5 The main problem. Tap all that apply, or Tap and speak, or Type instead.



6 Follow-ups. Questions chosen by the complaint ("Along with the fever…"); Skip is always allowed.



7 Review. The history read back; a red line when a danger sign was found; Change any answer.



8 Token. MED-016, the department, a QR to follow the turn; the kiosk resets for the next patient.



Who is answering (Marathi): the patient, a parent or guardian, or someone helping.



Online AI help is its own question; both answers look alike.



Today's token slip also carries the OPD card for next time.

Made for everyone

Voice guidance, buttons of at least 64 px, larger text and high contrast; **Call staff** on every screen; Back and Stop at any time. An idle kiosk warns, then clears itself after 60 seconds.

Answers are honest

"Don't know" and "Prefer not to say" stay open questions, never a "no". Voice keeps the patient's own words beside the English, with how sure the speech recognition was.

Same on a phone

The poster QR opens these screens on the patient's phone, with the same consent, safety rules and token; a reload or lost Wi-Fi resumes the same question.

Every question we ask

33 questions in 16 topics, each in Hindi, Marathi and English with one-tap answers, and open to speech or typing. A patient sees 8–15: the core three, up to two topics chosen from the complaint, and the background topics.

Before any question: the safety check

Chest discomfort · Trouble breathing · Heavy bleeding · Fainting or unable to stay awake · Sudden weakness or difficulty speaking · A seizure happening now. Any tap makes the visit urgent and alerts the staff.

ALWAYS: THE CORE THREE	
Main problem	What is the main problem that brought you here today? Fever, cough, difficulty breathing, stomach pain, vomiting or loose motions, headache, body or joint pain, skin rash or itching, burning while passing urine, weakness or tiredness, injury or fall — or in their own words.
Since when	Since when do you have this problem? Since today · 1–3 days · 4–7 days · 1–4 weeks · more than a month.
How bad	How bad is it right now? 0 means not at all, 10 means the worst. (Adds at most 4 points to the triage score.)

TOPIC (WHEN THE COMPLAINT MENTIONS…)	QUESTIONS (AT MOST TWO TOPICS A VISIT)
Fever fever, temperature	How is the fever? · Along with the fever, do you have any of these? (body ache, headache, cold, rash, burning urine, vomiting)
Cough, breathing breath, cough, wheeze, phlegm	Tell us about the cough or breathing. · Do any of these apply to you? (TB at home, asthma before, smoke, losing weight)
Stomach stomach, belly, vomit, loose motion, acidity	Do you have any of these stomach problems? · Do you have stomach pain? If yes, where?
Urine urine	Which of these do you have when passing urine? · Do you also have any of these? (fever, side pain, swelling of feet or face)
Pain pain, ache, sore, headache	Where is the pain? (skipped when stomach pain or headache is the only complaint) · What is the pain like?
Skin skin, rash, itch	What do you notice on your skin? · Where on the body is it?
Injury injury, fall, accident, fracture, sprain, bite	How did it happen? · Since it happened, do you have any of these? (cannot walk or lift, swelling, bleeding)
Tiredness tired, weak, fatigue, sleep, dizzy	Have you noticed any of these? (tired all the time, weight loss, poor sleep, dizziness, night sweats, feeling low)

BACKGROUND (EVERYONE)	QUESTION
Long-term illness	Do you have any long-term illness? (diabetes, high BP, heart disease, asthma, TB, thyroid)
Operations	Have you ever had an operation? (caesarean, appendix, hernia, gallbladder, cataract, bone or joint, other)
Pregnancy	Are you pregnant now? Only asked from 12 to 50 years, and not of men.
Medicines	Do you take any medicines regularly? You can also say their names.
Allergies	Are you allergic to any medicine or food?
Family	Does anyone in your family have any of these? (diabetes, high BP, heart disease, TB, cancer)
Habits	Do you use any of these? (smoking, chewing tobacco or gutkha, alcohol) — from 12 years.
Anything else	Is there anything else you want the doctor to know?

SWITCHED ON BY THE HOSPITAL	QUESTIONS
Review of systems	In the last month, have you had any of these? Tap all that apply. One screen covering 11 body systems; systems already asked about are left out.
Ayurveda (AYUSH OPDs)	How is your hunger? · How do you feel after eating? · How are your bowel movements? · How do you sleep? · Can you do your daily physical work? · What do you usually eat?

The trigger words exist in all three languages. With AI consent, up to three extra follow-ups may be written for the complaint; they are labelled "AI follow-up" for the doctor. Every question, option and rule is a readable file marked "draft — needs clinician review".

The OPD card, the patient's phone, and their rights

What the OPD card is

An **OPD number** (for example `26-000001`) and a **six-digit PIN**, printed on the token slip at the end of check-in. The number stays with the patient at that hospital; the PIN is worked out from a secret each time and never stored.

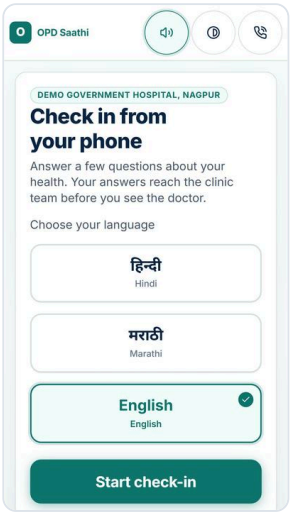
How it is used

- **At the kiosk:** "Yes, I have my OPD card", then the number and PIN. Today's age and sex must fit the record; wrong tries are limited and audited; the shared kiosk never shows what an earlier visit was about. The patient says whether today is a follow-up or a new problem.
- **At the desk:** a nurse searches by OPD number or phone (results show number, age, sex and visit count — no names), confirms in person, and links the visit.
- **Lost card:** the nurse re-issues it with a new PIN; the number and the records stay, and the old PIN stops working at once.
- **For the doctor:** the visit shows the patient's *Previous visits* (page 9).

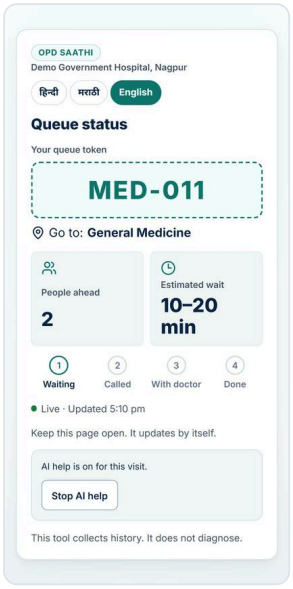
A printed PIN is not strong enough for patients to open their own records, so it only links visits (ADR 014).



A returning patient enters the OPD number and PIN from the last slip.



The waiting-hall poster's QR opens check-in on the patient's phone.



Live status: people ahead, a wait range, progress, "Stop AI help".



The privacy notice, a draft until counsel approves it.

Phone check-in

An administrator prints a *Phone check-in QR* (Settings → Kiosks). Patients answer the same interview with the same rules; staff see a **Phone** badge. "Replace QR" retires old posters; the hospital can switch it off.

The live status page

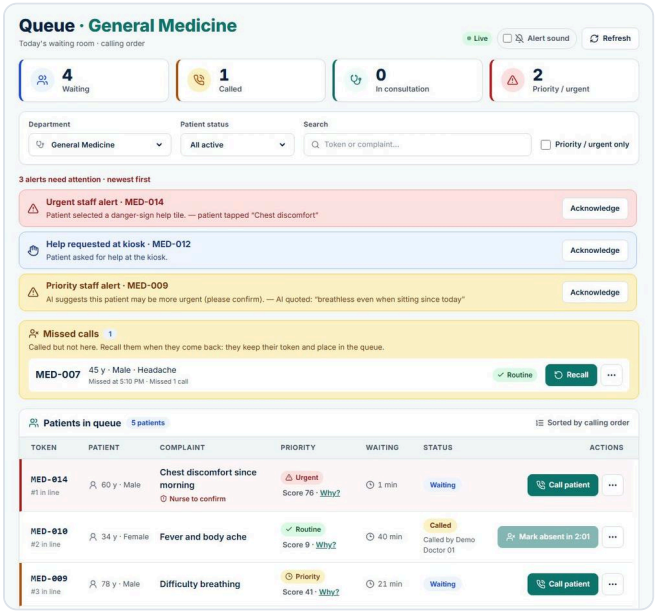
From the QR on the token: the token, people ahead, a five-minute wait range (never a falsely exact time) and Waiting → Called → With doctor → Done. The phone vibrates when called. No name or health details appear.

Rights and the notice

What is collected, why, who sees it, how long it is kept, the patient's rights and where to complain, in three languages, with the hospital's grievance officer and a printable poster. AI help and SMS can be stopped from the phone.

SMS and WhatsApp (token, then 5, 2 and 0 people ahead) are built but stay **off** until the hospital has a provider and registered templates (DLT). Only the patient's own tick asks for them.

The live queue



General Medicine: counters, three kinds of alert, the missed-calls panel and patients in calling order, with "Nurse to confirm", the Phone badge, "Called by" and a countdown before "Mark absent".

Live and reliable

- Changes reach every board within a second; a 5,000-patient list applies an urgent change in under 100 ms.
- A board whose sign-in ends is closed by the server; yesterday's open visits close themselves after midnight.

What is on the board

- **Counters** (waiting, called, in consultation, urgent) that filter; department, status and search filters.
- **Alerts**, newest first, each with Acknowledge: Urgent danger sign or red flag, Help requested, and Priority "AI suggests… please confirm", always quoting the patient. Unacknowledged urgent alerts sound again every 60 seconds.
- **Patients** in calling order: token, age and sex, complaint in English, priority with score and *Why?*, waiting time, status.
- **Next to call** with a one-click Call next.

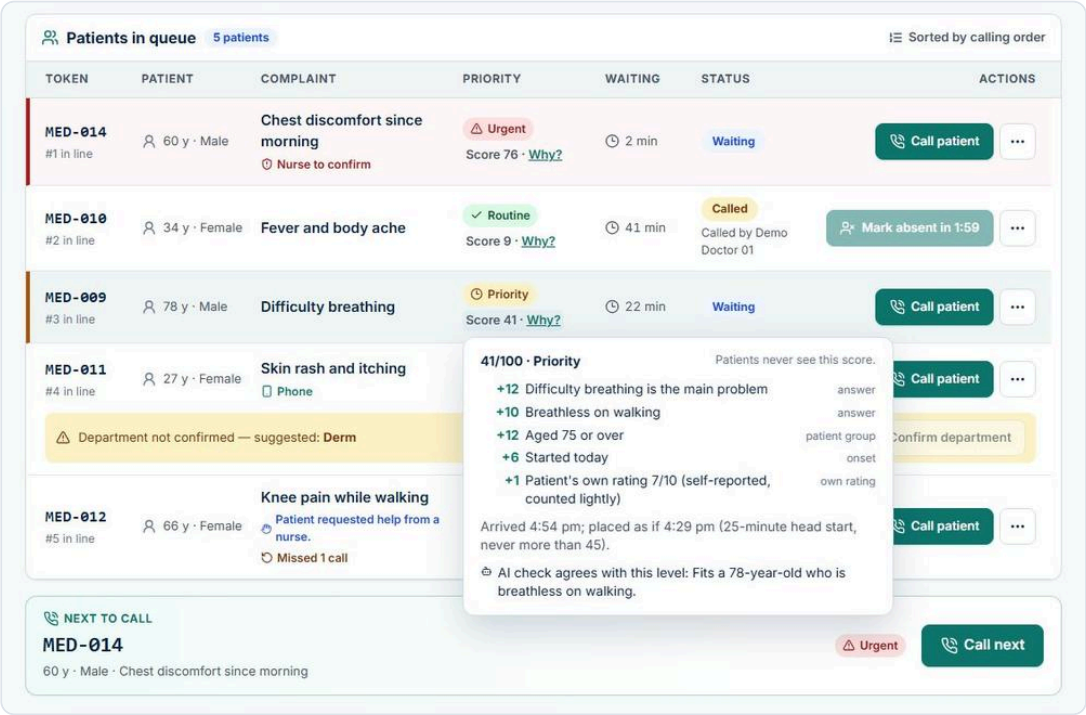
Calling, fairly and safely

- **One patient per doctor:** a doctor's call reserves the patient; two doctors pressing Call next together get two different patients.
- **Missed calls:** "Mark absent" unlocks after the hospital's grace period (3 minutes by default, counted down). The patient keeps token, urgency and arrival time; **Recall** puts them back in place; **Put back** undoes a wrong call; **No-show** only after a miss.
- **Nurse to confirm** stays on a self-reported danger sign until a nurse has seen the patient.
- Department and priority changes need a written reason; every action is checked against the person's department.
- Board in **Hindi and Marathi**; alerts and Call next work with **Tab and Enter** alone.

WHAT IT IMPROVES

The sickest patients surface in seconds; a patient who stepped out is not punished; two doctors never call the same person; and every call, absence and override says who did it.

How the fair queue works



"Why?" lists every point of a 41/100 Priority score, the head start it gave, and the AI check. Patients never see this.

The score (0–100), staff only

- Points come from objective signs: an urgent red flag or danger tile 70, a priority flag 30, warning answers up to 25, vulnerability (infant +20, aged 75+ +12, pregnant +12), started today +6.
- The patient's own 0–10 rating adds at most 4 — typing "10" cannot jump the queue.
- Routine under 30 · Priority 30+ · Urgent 70+ or any urgent red flag.

The calling order

- Urgent patients go first. Everyone else keeps arrival order with a **head start** of 0.6 minutes per point, never more than 45 — so nobody waits for ever.
- **AI is a checker, not a decider**: it may raise a "please confirm" alert, never lowers urgency, and alone counts as Priority at most.
- Weights and the 15 red-flag rules are readable files marked "draft — needs clinician review".

The Patients page: every visit, and help at the desk

Register

Every visit for today, 7 or 30 days, with status tiles, filters, search and "Open case sheet" — found in milliseconds even with years of visits.

Assisted check-in

A nurse checks in a patient who cannot use the kiosk; consent is still shown first and recorded as given at the desk, then the same interview, rules and queue follow.

Returning patients at the desk

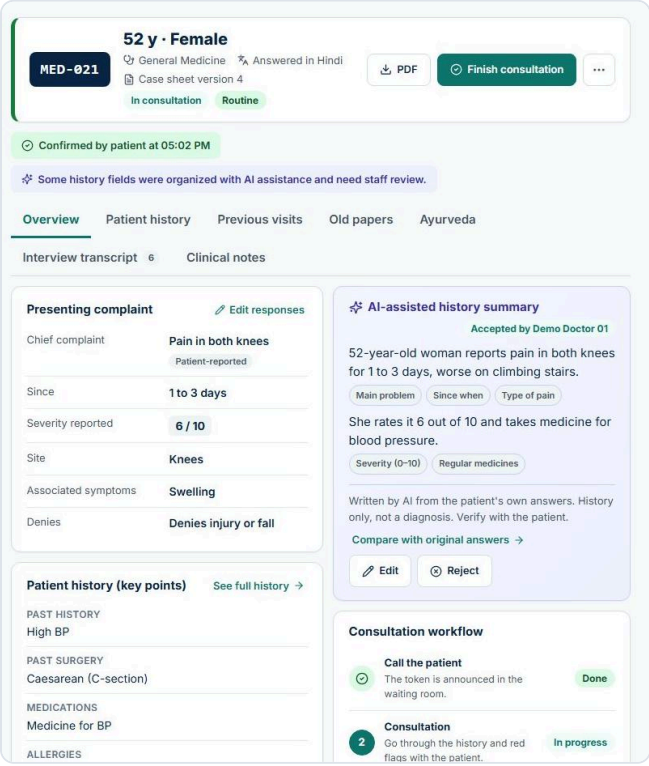
Search by OPD number or phone, confirm in person, then link; re-issue a lost card with a new PIN (page 6).

WHAT IT IMPROVES

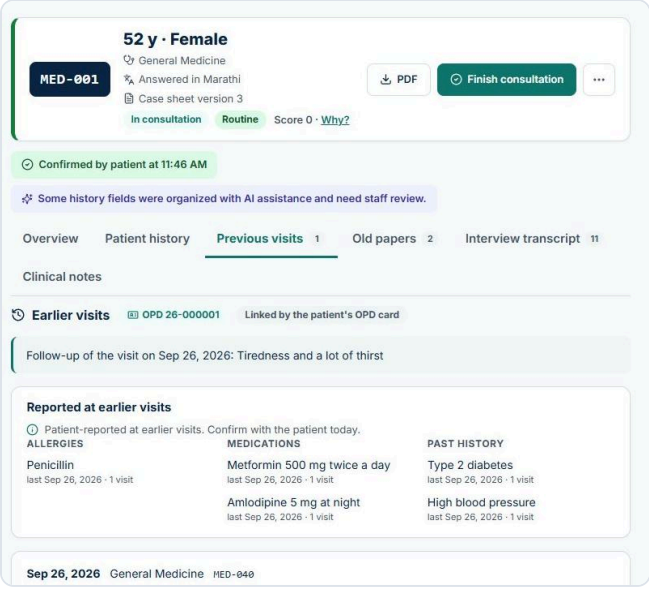
Anyone can ask "why was that person called before me?" and get a line-by-line answer; nobody is left out because they cannot use the kiosk or forgot their card.

The case sheet, with its sources

Left: today's patients. Right: one patient in tabs — Overview · Patient history · Previous visits · Old papers · Ayurveda · Interview transcript · Clinical notes — so urgent warnings and the AI draft to check come first and the details are one click away.



Each sentence of the AI summary carries chips for the answers it rests on; here the doctor has accepted it.



Previous visits of a patient linked by her OPD card: a follow-up of 26 September, with what she reported earlier.

Header

Token, age and sex, department, language, case-sheet version, status, priority with *Why?*, and one main action that follows the visit (Call → Start → Finish), plus PDF. "Answered by a parent or guardian" when someone else gave the history.

The AI draft: a sentence, its sources, a decision

- The model writes the brief as sentences that each cite the questions they rest on; the server drops any sentence citing nothing the patient was asked, and the output guard checks every one.
- A chip opens that answer in the transcript, highlighted; "Compare with original answers" does the same for the whole brief.
- The doctor records **Accept**, **Edit** (their words; the AI wording is kept for the record) or **Reject** — versioned and audited. The PDF names the reviewer and never prints a rejected draft.

Patient history and the transcript

Every field shows the English text, the patient's own words and how they gave it (tap, voice with confidence, typed). Edits create versions marked **Staff-verified**; two people editing at once cannot overwrite each other. The transcript of every question and answer cannot be changed.

Previous visits

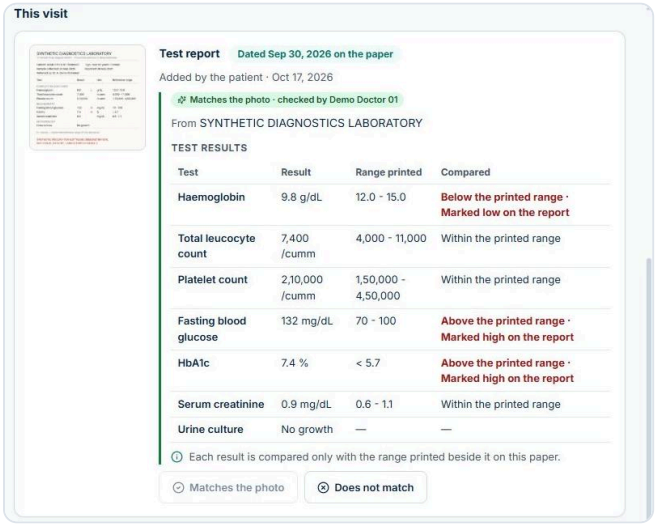
Up to 20 earlier visits at the same hospital: date, department, complaint, red flags and the reviewed AI brief, plus allergies, medicines and conditions with the date each was last reported — labelled "patient-reported, confirm today". Doctors and nurses see it only while the patient is under their department's care today.

Visit timeline

Every change to the visit, who made it and when: token issued, called, marked absent, recalled, consultation started. The application can add lines but never change or delete them.

> **WHAT IT IMPROVES**
> The doctor starts with a checked summary instead of a blank page, sees exactly which answer each AI sentence came from, and knows the patient's past without asking again.

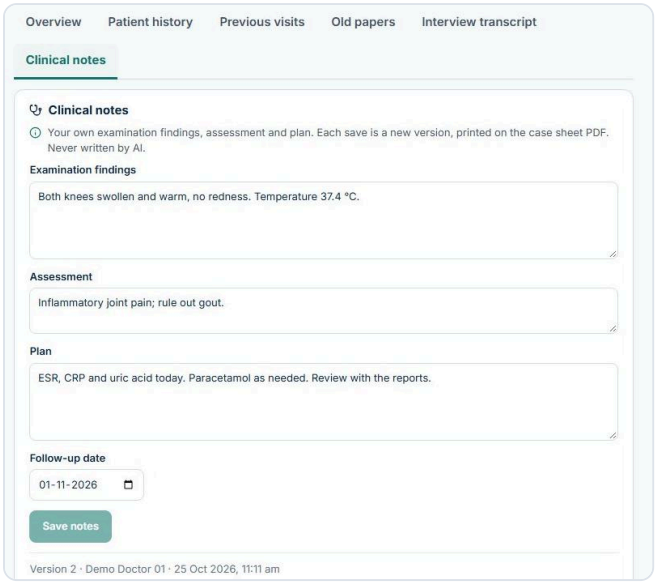
Old papers, results over time and the doctor's own notes



A test report beside what AI read from it, accepted by the doctor; each result compared only with the range printed on that paper.



Haemoglobin across three visits' reports, oldest first; a paper still awaiting a decision is counted, not shown.



Clinical notes: the doctor's own findings, assessment, plan and follow-up date; each save is a version.

Old papers

- Photos from the kiosk, and up to 10 a visit added by staff, newest first by the date printed on them; earlier visits' papers while the patient is under care.
- With AI consent, each paper gets a reading: source, diagnoses and medicines as written, results with value, unit and the printed range.
- "Below / Within / Above the printed range" is plain arithmetic against **that paper's own range**, plus the lab's own high/low mark — never a general range, never the model's opinion.
- The doctor marks **Matches the photo** or **Does not match**. Without AI consent the photo is kept unread. Every view of a photo is audited.

Results over time

Only readings a doctor accepted, from this and earlier visits, one table per test, each result next to the range printed on its own paper. A trend that would take minutes to find in a pile of papers is one glance.

FHIR export

A visit can be exported as a FHIR R4 bundle (patient, encounter, complaint and conditions as unconfirmed, allergies, medicines as stated, accepted results, the interview, consents and who answered) through the API, for connecting to a hospital system. It is marked "not ABDM-certified".

The doctor's own notes

- Examination findings, assessment, plan and a follow-up date, written by the doctor and **never by AI**.
- Each save is a new version; earlier ones stay readable; a save that does not start from the latest version is refused, so two doctors cannot overwrite each other.
- The latest version is printed on the case-sheet PDF.

PDF and speed

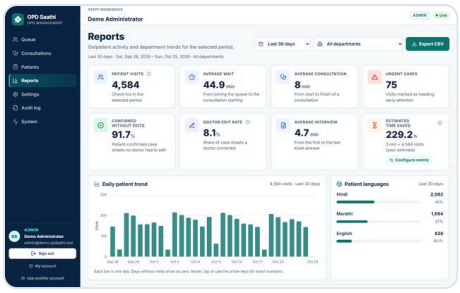
PDF / Print always gives the full history-only case sheet in three scripts, with the reviewed AI summary, open questions and Ayurveda findings. A whole visit — call, start, accept the draft, write notes, finish, print — takes **6 clicks and one key**.

MAP OF THE APP

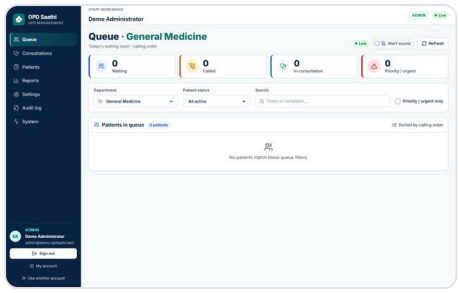
Every screen, and who uses it

One web app serves every screen. In the demo the addresses start with `http://localhost:8080`; staff sign in at `/staff` and see only their role's pages.

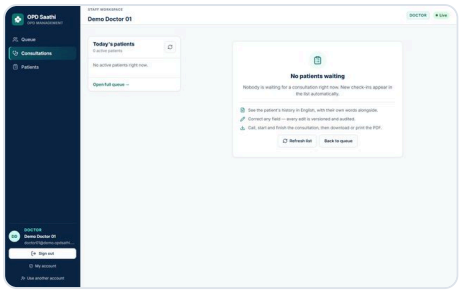
SCREEN	WHO	ADDRESS	WHAT IT SHOWS AND DOES
Kiosk check-in	Patients	`/kiosk`	The interview from language to token (page 4); paired once with an activation code.
Phone check-in	Patients, from the poster	`/m/<code>`	The same interview on the patient's own phone.
Live status	Patients, from the token QR	`/s/<token>`	Place in line, wait range, called; stop AI help or SMS.
Privacy notice	Anyone	`/privacy/<hospital>`	What is collected and why, the patient's rights, the grievance officer.
Queue	Nurse, doctor, admin	`/staff/queue`	The live waiting room: counters, alerts, calling order, Call next, missed calls.
Consultations	Doctor, admin	`/staff/consultations`	Today's patients and the case sheet: AI draft, history, previous visits, papers, transcript, notes, PDF.
Patients	Nurse, doctor, admin	`/staff/patients`	Register of visits, assisted check-in, finding returning patients, re-issuing cards.
Reports	Admin	`/staff/reports`	The dashboard: visits, waits, consultations, urgency, accuracy, languages, departments, CSV.
Settings	Admin	`/staff/settings`	Staff accounts, kiosks and phone QRs, departments, hospital switches.
Audit log	Admin	`/staff/audit`	Who did what, when, to which record; search and export.
System	Admin	`/staff/system`	Health of the database, queue store, worker, backups, certificate, AI, voice and each kiosk.
My account	Everyone	`/staff/account`	Password, two-step sign-in, own sessions.



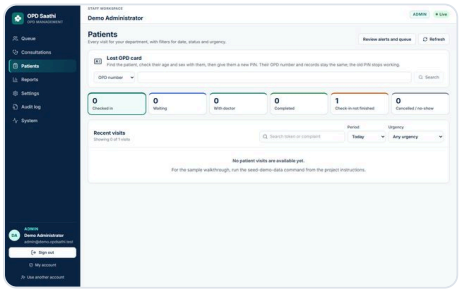
Reports — 30 days of the demo hospital.



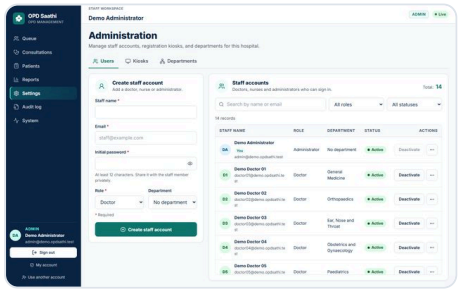
Queue — empty until someone checks in.



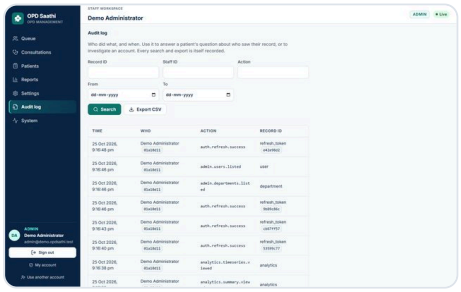
Consultations — the doctor's list.



Patients — the register and desk tools.



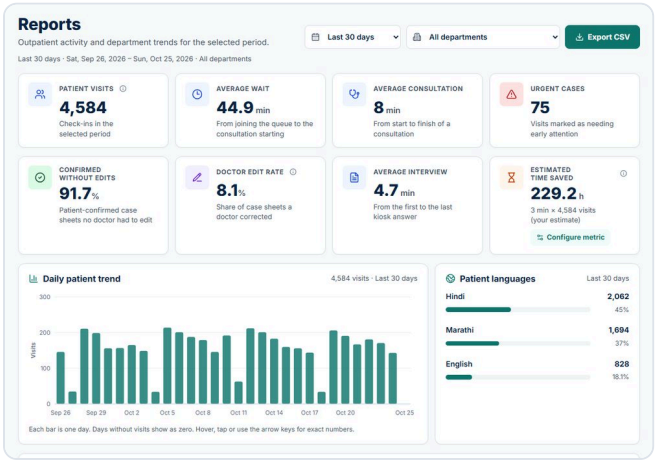
Settings — users, kiosks, departments.



Audit log — every view and change.

Today's demo, 25 October, with the new logo in the sidebar. Add `?demo=1` to any address for a presentation ribbon (kiosk) or a "How it works" panel (staff).

Reports, settings, the audit log and the System page



Reports for 30 days of the demo hospital: eight cards, the daily trend and the language mix.

Reports

For today, 7, 30 or 90 days and any department: patient visits, average wait, average consultation, urgent cases, **confirmed without edits** (how often the kiosk history was right), doctor edit rate, average interview, and an estimated time saved (labelled, with minutes per visit set by the administrator). Daily trend, languages, department load and a CSV export protected against spreadsheet formulas. Every view is audited.

Counted one day at a time

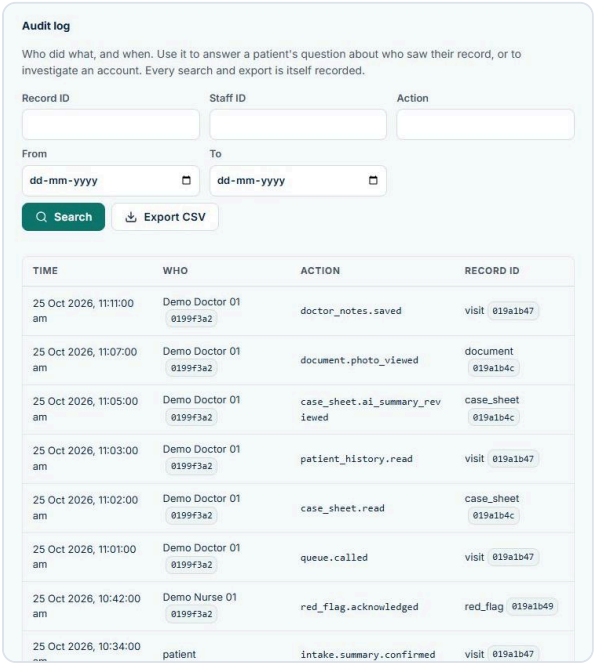
The figures come from one row per department and day. The worker recounts today and yesterday every 5 minutes (0.1–0.2 s at a million visits) and every day at 03:07, so a late change to an older visit shows by the next morning. Before 25 October the whole history was recounted every 5 minutes (10 s at a million visits).

Settings

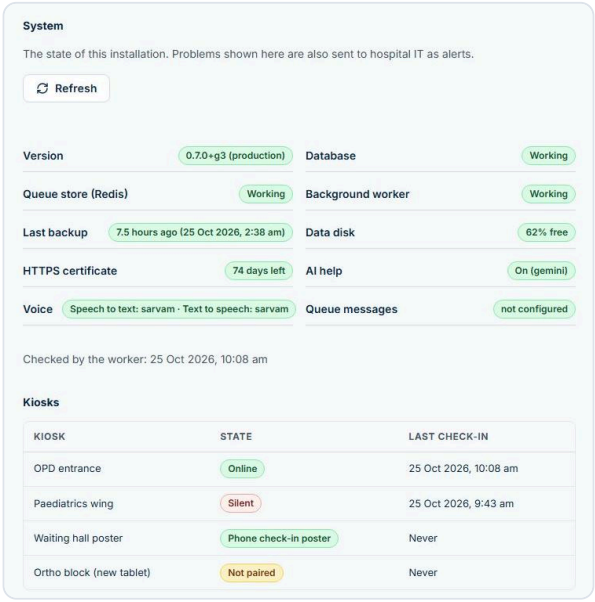
- **Users:** create, edit, deactivate; reset a password (changed at first sign-in), unlock, sign out everywhere; two-step sign-in required for administrators.
- **Kiosks:** a one-time 24-hour activation code; Online / Offline / Not seen yet; secrets renew every 30 days; phone check-in QRs to print and replace.
- **Departments:** code, token prefix (MED → MED-016), names in three languages.
- **Hospital switches:** call grace period, Ayurveda, review of systems, online AI, SMS, phone check-in — each audited; risky ones ship off.

Audit log and System

- **Audit log:** every sign-in, every view of a case sheet, photo or report, and every change — who, when, which record, never answers or names. Search by record, person, action and dates; export CSV. Nobody can change it.
- **System:** version; database, queue store and worker; last backup, disk, certificate; how AI, voice and SMS are set up; each kiosk's last check-in. The same problems reach hospital IT as e-mail alerts.
- **Printables:** staff cards, patient signs, the privacy poster and a paper fallback pack.



Audit log: a visit's trail, from the patient's consent to the doctor's notes.



System: one look at the installation's health.

AI as an assistant, and life without it

AI needs three yeses: the server has a key, the hospital has switched it on, and the patient agreed. Without any of them, every core feature still works on the hospital's server.

FEATURE	WITH AI (GEMINI; SARVAM FOR INDIAN VOICES)	WITHOUT AI, OR WHEN THE PATIENT SAYS NO
English case sheet	Free text translated; a short doctor's brief whose every sentence cites its answers, for the doctor to accept, edit or reject	Tapped answers translated from the bilingual question bank; free text stays in the patient's words
Questions	Up to three follow-ups written for this complaint, in three languages, labelled "AI follow-up"	The fixed, logical question flow (page 5)
Voice	Server speech recognition (Sarvam, then Gemini) and natural voices	Recorded question audio and the device's own voice; tapping and typing
Old papers	Each photo read as a draft for the doctor to compare	The photo is kept unread for the doctor to read
Safety	Rules, plus an AI second opinion that can only raise urgency and must quote the patient	The rules engine (15 draft rules) and danger tiles
Routing, queue, tokens, reports	Always rule-based and the same with or without AI	

Guard-rails on every AI answer

- An output guard rejects text that diagnoses, advises treatment, names doses or contains links (100 of 100 adversarial prompts blocked in testing).
- Names and phone numbers are removed before anything is sent; AI red flags must quote what the patient said; AI never lowers urgency.
- Drafts are labelled until a doctor reviews them; a rejected draft is never printed.
- If an AI service is slow or down, the kiosk quietly continues with the question bank.

Why this design

A hospital can start with no AI cost and no data leaving the building, and switch AI on only when its contract and its patients allow. Safety never depends on AI being right: rules decide, AI only adds caution, people confirm. A patient who said yes can withdraw from their own phone; from then on nothing more about the visit goes to AI services.

In today's demo check, Gemini (gemini-3.5-flash-lite) answered in 2.4 s and the Sarvam voice in 0.8 s.

Safety, privacy and security

Clinical safety

- History only — no diagnosis, disease names, medicines or advice anywhere, in any output.
- Danger signs first; uncertainty escalates, never lowers urgency; lowering needs a recorded staff decision.
- The patient confirms the read-back; doctors correct anything; every change is versioned.
- Red-flag rules, triage weights and wording are drafts with a hazard log; production refuses to start with unsigned clinical content when the hospital requires sign-off.

Privacy (DPDP Act 2023)

- Consent by purpose: care (required), online AI help and SMS (optional, each with its own notice, withdrawable).
- A full privacy notice in three languages and a grievance officer.
- Names, phones, answers, case sheets, notes and photos encrypted in the database; only age and sex required.
- Status pages show no personal details; a retention tool removes or pseudonymises old records once the hospital approves the periods.

Security

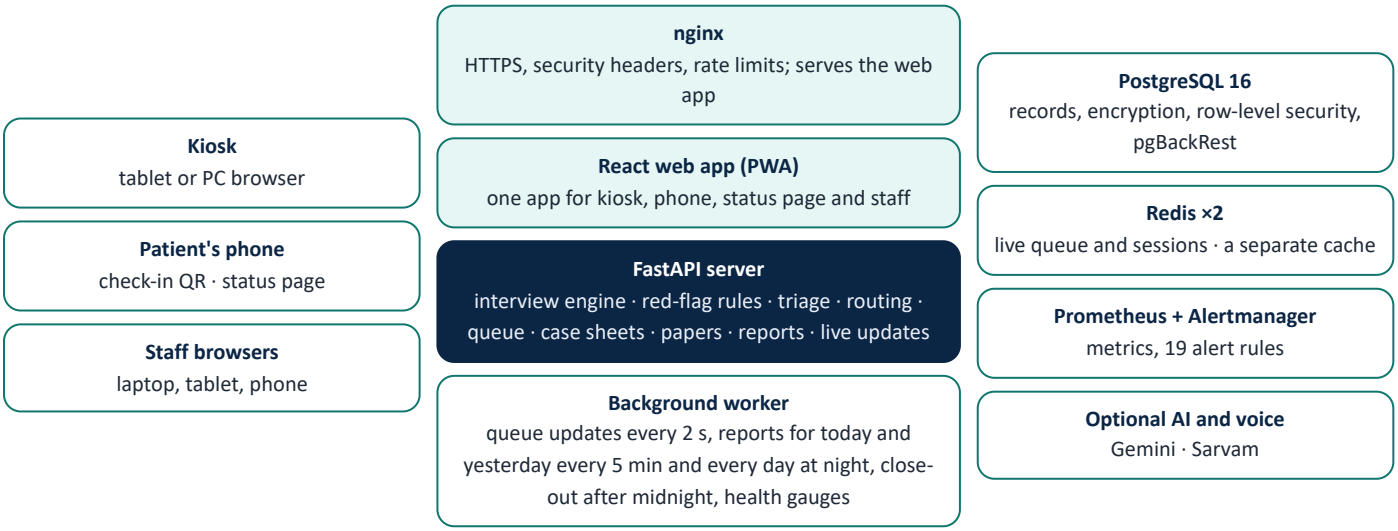
- Each hospital's rows walled off by the database itself (row-level security), not only by code; least privilege for every service.
- Two-step sign-in, password reset with forced change, sign-out everywhere; Argon2 password hashes; lockout and rate limits.
- Kiosks paired with one-time codes; their secrets and the server's keys can be rotated.
- HTTPS only, strict browser security headers, an append-only audit log with its own screen.
- On every change: image and code scanners (Trivy, Semgrep, Bandit, Gitleaks, dependency audits); nightly attack scans (OWASP ZAP) and API fuzzing; a threat model and an OWASP ASVS Level 2 self-assessment.

Reliability

- Every request carries a unique key: a double tap or a retry never creates two visits or two calls.
- Two people editing one case sheet or one set of notes cannot overwrite each other.
- If the queue store is lost, the queue is rebuilt from the database; the worker survives a Redis outage.
- Backups to the minute with a restore drill; 19 tested alerts reach a person by e-mail; failure drills run every night.

For real patients, a penetration test and clinical and legal sign-off still have to happen (page 19). The engineering for them is ready; the decisions belong to people.

How it is built



LAYER	TECHNOLOGY
Web app	React 19 and TypeScript 5.9, Vite 7, React Router 7, TanStack Query, Zustand, Tailwind CSS 4, i18next (Hindi, Marathi, English), lucide icons, an installable PWA with an encrypted offline form, virtualised lists for very long queues
API	Python 3.12, FastAPI, Pydantic, SQLAlchemy (async) with asyncpg, Gunicorn and Uvicorn workers, WebSockets and server-sent events for live updates, WeasyPrint for PDF case sheets, structlog, PyJWT, Argon2, pyotp (two-step sign-in), the Google Gen AI SDK
Data	PostgreSQL 16 with row-level security and Alembic migrations (35 so far); Redis 7 for queue state and a separate cache; ARQ for background jobs; AES-GCM field encryption with rotatable keys
AI and voice	Google Gemini (summaries, translation, follow-ups, reading papers, a red-flag second opinion); Sarvam (Indian-language speech recognition and voices); the browser's own voices as a fallback
Running it	Docker Compose with a production overlay (TLS in nginx, secrets as files, non-root containers), pgBackRest, Prometheus and Alertmanager; images pinned by digest, signed with cosign, with a component list
Quality gates	GitHub Actions on every change: backend, frontend, production smoke test with browser tests against the real API, image scans and security — all must pass before code reaches `main`; nightly load (k6), real installation, upgrade, failure drills, OWASP ZAP and Schemathesis
Tests	pytest with Hypothesis, Vitest with a coverage gate, Playwright with axe accessibility scans at seven screen widths and three languages

WHAT IT IMPROVES

An ordinary, well-supported open-source stack that a hospital's IT can run on one server, with every decision written down as an ADR in `docs/DECISIONS.md`.

9 · EVIDENCE

Tests and results

1,087
API tests passed
on real PostgreSQL and Redis (3 skipped), 25 October; migrations run up, down and up

155
browser tests
kiosk, phone and staff screens; ten full runs in a row with no retry

137
web unit tests
with a coverage gate that can only go up

13 / 13
CI checks green
on the last change (#74): backend, frontend, security, images, smoke, install, upgrade, load

Synthetic evaluations

CHECK	RESULT
Red flags that must be caught (held-out set)	60 / 60
Negations not falsely flagged ("no chest pain")	50 / 50
Department routing (held-out set)	96 / 96
History extraction with zero invented fields	100 / 100
Adversarial prompts blocked by the AI output guard	100 / 100
Urgency never decreases when a warning sign is added	500 / 500
Golden interviews completed correctly (three languages)	30 / 30

Speed and scale

- **Load at a district hospital's peak** (300 check-ins an hour, 20 boards, 200 status pages, 8 doctors, 15 minutes): 18,102 requests, 0.02% failed, no token issued twice; 95th percentile check-in step 29 ms, staff board 16 ms, call next 165 ms.
- **A million visits** (400,000 patients, 11.5 million answers): the screens staff keep open answer in 9–12 ms after five indexes (migration 0034); query-count tests stop any screen from running one query per row.
- **Reports per day** (migration 0035): the 5-minute refresh fell from 10 s to 0.1–0.2 s, with numbers identical to before, row for row.
- Danger-tile alerts reach staff boards in under a second; a 5,000-patient board applies an urgent change in under 100 ms.
- **Accessibility:** no serious or critical issues in automated scans; Lighthouse accessibility 100 on kiosk and status pages; no sideways scrolling from 320 to 1600 px.

Today's demo check (25 October)

Six services healthy; the web app and API answer; the database at the latest migration (0035); Gemini answered in 2.4 s and Sarvam in 0.8 s; the returning patient and her old report ready — "Ready for the demo". Every staff screen opened under an administrator and a doctor account.

> These are synthetic tests: they prove the software behaves as designed, not clinical accuracy. Clinician and native-speaker review is still required for real patients.

What was built, and when

From an empty repository on 5 October to a pilot-ready system on 25 October: 26 pull requests merged, and since 10 October every change has reached `main` through a pull request that passed CI.

WHEN	WHAT WAS BUILT
5–7 October Build phases 0–14	Project set-up; staff sign-in and roles; the data model with each hospital walled off; the kiosk interview and question bank in three languages; tokens, the queue and the live status page; danger signs, red-flag rules and department routing; nurse, doctor and administrator screens with the PDF case sheet; voice answers (Sarvam); Gemini with safety guard-rails; the read-back with voices; queue messages; the Reports dashboard; the offline kiosk form; security hardening; a demo mode and faster loading.
10–11 October Pull requests #6–#10	Back and quit on the kiosk and fixes to the question logic; staff screen layout; data quality in reports; kiosk activation, replacement and deletion; hospital set-up; staff sessions and alerts.
16–17 October IIT Bombay readiness, #11–#16	Phone check-in on the hospital Wi-Fi; the OPD card, returning patients and previous visits; the fair queue (missed calls, recall, one patient per doctor, abuse limits); AI summaries with sources; old papers read as drafts; consent by purpose; demo data and rehearsal tools; the Phase 7 plan and the 83-page documentation.
23 October Phase 7 G0, G1, G2: #17, #18, #35	**G0** fixes found by a code review (SMS only by the patient's tick, worker keys). **G1 secure build:** HTTPS, two-step sign-in, staff account tools, the audit log screen, key rotation, security scans. **G2 operable:** 19 alerts, backups with restore drills, the System page, load tests, signed releases, the paper fallback pack.
24–25 October Phase 7 G3: #59, with dependencies #36, #48, #60	**Pilot-ready engineering:** the doctor's own notes, review of systems, results over time, lost OPD card, the AI switch, who is answering, the privacy notice, FHIR export, the million-visit index review, ten clean browser runs in a row; React 19 and dependency updates tested together; the first release, `v0.7.0` (24 October).
25 October #72, #73, #74	The team's feature guide; **the OPD Saathi logo** (a plus that speaks, with a voice wave: care, conversation, voice) in the tab, home screen, loading screen, staff sidebar and kiosk; **reports counted per day**.

Original features

Kiosk interview, voice, three languages, danger signs, red-flag rules, routing, triage, live queue, status page, case sheet, PDF, reports, offline form, AI with guard-rails.

Added for IIT Bombay

Phone check-in, OPD card, previous visits, fair queue rules, AI with sources, old papers, consent by purpose.

Added in Phase 7

Security, operations and the G3 features: notes, review of systems, results over time, privacy notice, audit log and System screens, the logo and per-day reports.

Running the demo

Start, check, stop

1. **Start** Docker Desktop, then in PowerShell at the repository root: `.\start-demo.ps1 -WebPort 8080`. The first run builds the images (several minutes). It migrates the database, prepares the demo hospital, 30 days of history and the returning patient, prints the links and accounts, and opens the kiosk.
2. **Check**, 30 minutes before: `.\scripts\demo-check.ps1` — services, web and API, database, a live AI and voice request, demo patients. It ends with "Ready for the demo".
3. **Phones:** `.\scripts\lan-check.ps1` when phones on the same Wi-Fi cannot open the QR.
4. **Stop:** `docker compose down` (the data is kept).

Accounts and demo data (fictional)

- **Administrator:** `admin@demo.opdsathi.test`; password in `.env` as `DEMO_ADMIN_PASSWORD`.
- **Doctors** `doctor01` – `doctor08@demo.opdsathi.test`, one per department (doctor01 is General Medicine); **nurses** `nurse01` – `nurse05`; password `DEMO_STAFF_PASSWORD`.
- **Returning patient:** the start script prints her OPD number and PIN (today `26-000001` / `305179`); at the kiosk answer age 52, Female, then "Yes, I have my OPD card". Her old test report to photograph: `docs/demo-papers/synthetic-lab-report.png`.
- **Presentation mode:** add `?demo=1` to the kiosk or staff address.

An eight-minute run-through (full script: `docs/DEMO.md`)

TIME	SCREEN	DO, AND THE POINT IT MAKES
0:00	Kiosk	Choose मराठी; agree to the care notice; say yes to AI help — a separate, optional choice.
0:55	Kiosk	Age 52, Female; "Yes, I have my OPD card" with the number and PIN; "the same problem" — her record is linked but never shown on the kiosk.
1:40	Kiosk	Safety screen "No, none of these" — danger signs come first.
1:55	Kiosk	Answer by tapping, and once by Tap and speak in Marathi; on Review hear the read-back and photograph the old test report; confirm.
4:00	Kiosk + phone	Token, department and QR; scan it: the live status page with "Stop AI help".
4:30	Nurse (Queue)	Her token, English complaint and wait time, urgency first.
5:00	Doctor	Call, Start; the AI brief with source chips; Accept. Previous visits; Old papers with the printed-range flags; Matches the photo; Finish; PDF.
8:00	Admin	Reports (30 days of history); Settings → Kiosks → Show QR for phone check-in.

Optional: an emergency in one minute

A new check-in; on the safety screen tap Chest discomfort. The nurse's board shows an urgent alert at once; after confirmation the patient is first in line, marked "Nurse to confirm".

Do not promise

Real SMS (it is off; patients see the token on screen and on their phone), ABHA/ABDM integration, or any diagnosis. If AI fails, say so at the AI step: every step has a fallback.

Limits, what is left, and the words used

Boundaries on purpose

- No diagnosis, disease names, medicines, doses or drug-interaction checks (ADR 015).
- It knows only what the patient said: the history is their own account, translated, not verified.
- Patients see their place in the queue, not their records (a printed PIN is not strong enough sign-in; ADR 014).
- One server per hospital, inside its network; not built to face the internet.

Before real patients (owned by people, not code)

- A penetration test by an outside auditor.
- Clinicians' sign-off of questions, rules and weights; native speakers' review of Hindi and Marathi.
- Counsel's sign-off of the notice, consent texts, contracts, retention and breach procedure (DPDP Act).
- A usability test with real staff and patients; real devices over HTTPS on the hospital network; staff training.
- Waiting on outside parties: an SMS provider and DLT templates, more languages, staging hardware, ABDM.

Words used in the app

TERM	MEANING
Token	The patient's number for the day: department prefix and counter (MED-016 = the 16th General Medicine patient today).
OPD card	OPD number (for example 26-000001) and a six-digit PIN on the token slip; links the next visit to this one. The PIN is never stored.
Danger tile, red flag	A danger tile is one of six safety buttons on the kiosk; a red flag is a warning raised by a safety rule (or an AI second opinion) from what the patient said. Neither is a diagnosis.
Triage score, head start	A 0–100, staff-only score; each point places the patient 0.6 minutes earlier than their arrival, at most 45 minutes. Urgent patients go first.
Grace period, recall	The minutes after a call before staff may mark a patient absent; a recall puts them back on their original arrival time.
AI draft, sources	The AI's summary for the doctor, each sentence linked to the answers it rests on; the doctor accepts, edits or rejects it.
Printed range	The normal range printed beside a result on the patient's own paper — the only range results are compared with.
Consent purpose	Care (required), online AI help and SMS (optional), each with its own notice and record, each withdrawable.
Case sheet, transcript	The organised English history for the doctor, and the unchangeable record of every question and answer.
Activation code, check-in QR	A one-time, 24-hour code that pairs a kiosk; a printed QR that opens phone check-in and can be replaced at any time.
Row-level security	A database feature that makes each hospital's rows invisible to every other hospital.
FHIR	The international standard format for exchanging health records; OPD Saathi exports a visit as a FHIR R4 bundle.

Where to read more

`docs/DEMO.md` (demo script) · `docs/DEPLOYMENT.md` (installing in a hospital) · `docs/ARCHITECTURE.md` · `docs/SAFETY.md` and `docs/CLINICAL_SAFETY.md` · `docs/PRIVACY.md` · `docs/THREAT_MODEL.md` · `docs/EVALUATION.md` · `docs/SCALING.md` · `docs/LIMITATIONS.md` · `docs/DECISIONS.md` (ADRs) · `CHANGELOG.md`.